$zz^* = a^2 + b^2$

$\mathbf{q} = q_0 + q_1\mathbf{i} + q_2\mathbf{j} + q_3\mathbf{k}$

z

$z = r\,e^{i\theta}$

$|z| = \sqrt{a^2 + b^2}$

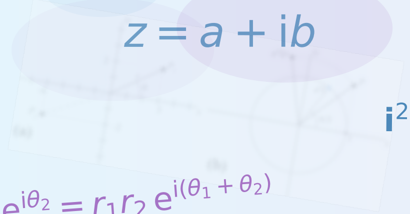
$|z| = \sqrt{a^2 + b^2}$

$\mathbf{q} = q_0 + q_1\mathbf{i} + q_2\mathbf{j}$ +

$e^{i\theta}$

i

$z = a + ib$



$r_1e^{i\theta_1}r_2e^{i\theta_2} = r_1r_2\,e^{i(\theta_1 + \theta_2)}$

$z = r\,e^{i\theta}$

$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -1$

$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}$

$r_1e^{i\theta_1}r_2e^{i\theta_2} = r_1r_2\,e^{i(\theta_1 + \theta_2)}$

q

$e^{i\theta} = \cos\theta + i\sin\theta$

$e^{i\theta} = \cos\theta + i\sin\theta$

$z = a + ib$

$zz^* = a^2 + b^2$